

MAN00035C



**School for Business
and Society**

Quantitative Analysis in Business Studies

2026 - Week 5

Week 5: Hypothesis Testing

Last topic

We looked at:

- Population vs Sample
- Types of sampling
- Inference **Population Parameter** via **Sample Statistics**
(μ vs $\bar{x}$ and σ vs s)
- Point Estimation vs Interval Estimation (confidence interval)
 - no 100% confidence
 - the Central Limit Theorem
 - t vs normal distribution
 - calculating sample size

This topic

We will look at Hypothesis Testing:

What happens if we know the value of some population mean or population proportion, and a sample taken from this population doesn't appear to agree with it?

Either

- The population parameter is not what we think

OR

- The difference between the sample estimate and population parameter is simply due to chance effects

Structure of today's session

We will learn about:

- Null hypotheses and alternate hypotheses
- Two approaches: Critical value approach and p-value approach
- Type I and Type II error

Reading: Oakshott Ch9, Taylor Ch7

What is a hypothesis?

A statement, claim or assertion about one or more populations.

e.g.

“The mean weight of a can of beans is 420g”

“50% of people take out holiday insurance”

To test these hypotheses, we set up a Test of hypothesis.

The hypothesis we want to test is called **null hypothesis**, which is written as H_0 .

e.g. $H_0 : \mu = 420 \text{ g}$

Or $H_0 : \pi = 50\%$

The alternative hypothesis

We want to see if we can disprove the null hypothesis.

- If we **reject** the null hypothesis, we automatically **accept** the alternative hypothesis, H_1 .e.g.,

H_0 : The mean weight of a can of beans is 420g

H_1 : The mean weight of a can of beans is not 420g

- This hypothesis can be ***two-tailed*** or ***one-tailed***:

H_1 : The mean weight of a can of beans is less than 420g

Stating the hypothesis

Say we have one sample...

	two sided alternative	one (upper) sided alternative	one (lower) sided alternative
Null hypothesis	$H_0 : \mu = \mu_0$	$H_0 : \mu = \mu_0$	$H_0 : \mu = \mu_0$
Alternative hypothesis	$H_a : \mu \neq \mu_0$	$H_a : \mu > \mu_0$	$H_a : \mu < \mu_0$

Example of two sided test
for mean

$$H_0 : \mu = 420 \text{ g}, \quad H_1 : \mu \neq 420 \text{ g}$$

Example of one sided test for
proportion

$$H_0 : \pi = 50\%, \quad H_1 : \pi > 50\%$$

Why we need a test statistic?

If $H_0 : \mu = 420$, and $\bar{x} = 400$. What can we conclude?

- Even if H_0 is true, $\bar{x}$ may differ from μ .
- This difference is due to random sampling variation.
 - The larger σ or the smaller n is, the more variable $\bar{x}$ is.

We need a way to judge whether $\bar{x} = 400$ is close to or far from 420.

H_0 cannot be an inequality:

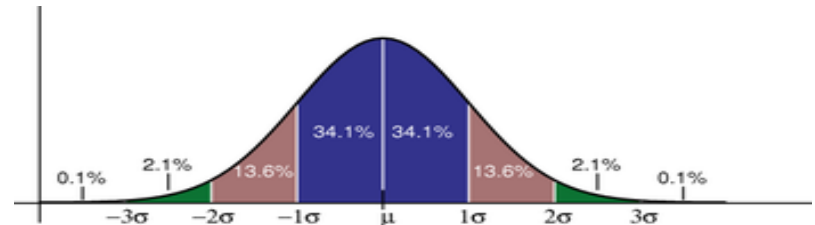
- We cannot calculate a statistic from many values.

Test statistic

A test statistic is

- a single numerical value calculated from sample data
- used to measure how closely the observed data matches the null
- Incorporating the impact of σ and n
- has a known distribution under null (also called **null distribution**)

An Example: $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ has a standard Normal distribution.



Two approaches to arrive at the decision

- In order to decide whether to reject the null hypothesis, a **test statistic** is calculated.
- The decision is made based on the numerical value of that test statistic.
- There are two approaches how to arrive at that decision:
 - the ***p-value approach***.
 - The ***critical value approach***.

p-value approach (1)

- Assume the Null Is True
- Ask: How Likely Is Our Data Under H_0 ? This is the **p-value**.
 - To calculate this, we need a specific value of the population parameter, e.g., $\mu = 420$ g.
- The p-value is the probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct.
 - Ranging from 0 to 1
 - A lower p-value indicates the findings are likely not due to random

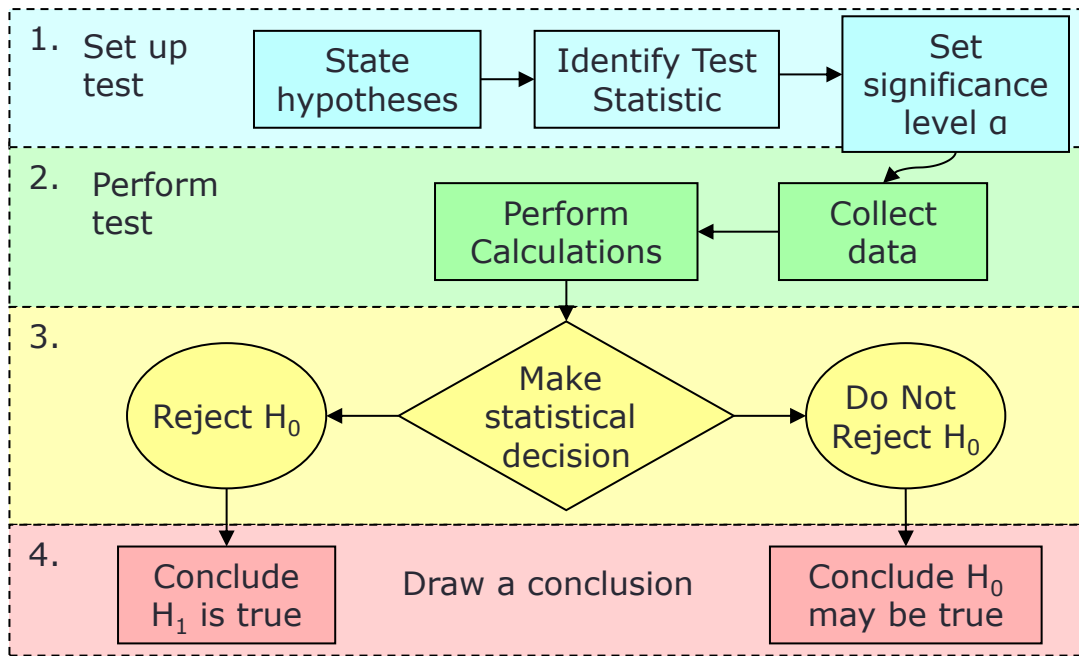
p-value approach (2)

- Compare p-value with a predetermined **significance level** α (usually 5%)
 - If p-value < significance level, we conclude that the data are unlikely under H_0 . Therefore, we reject H_0 and accept H_1 .
 - If p-value > significance level, we conclude we fail to reject H_0 ,
 - meaning we do not have enough evidence to reject H_0
 - Not meaning H_0 is true. (Inaccurate to say we accept H_0)

Critical value approach

- The critical value (CV) is computed based on the chosen significance level and the distribution of the test statistic.
- Compare the value of the test statistic with the CV.
- Relationship between the two approaches:
 - The test statistic $\leftrightarrow$ p-value
 - The critical values $\leftrightarrow$ significance level
- Both approaches lead to the same conclusion.
- Calculating CV is simpler as it does not rely on the random samples.

Steps for carrying out a hypothesis test



Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally, 5%), null distribution, and determine the critical values

Step 3. Calculate the test statistic

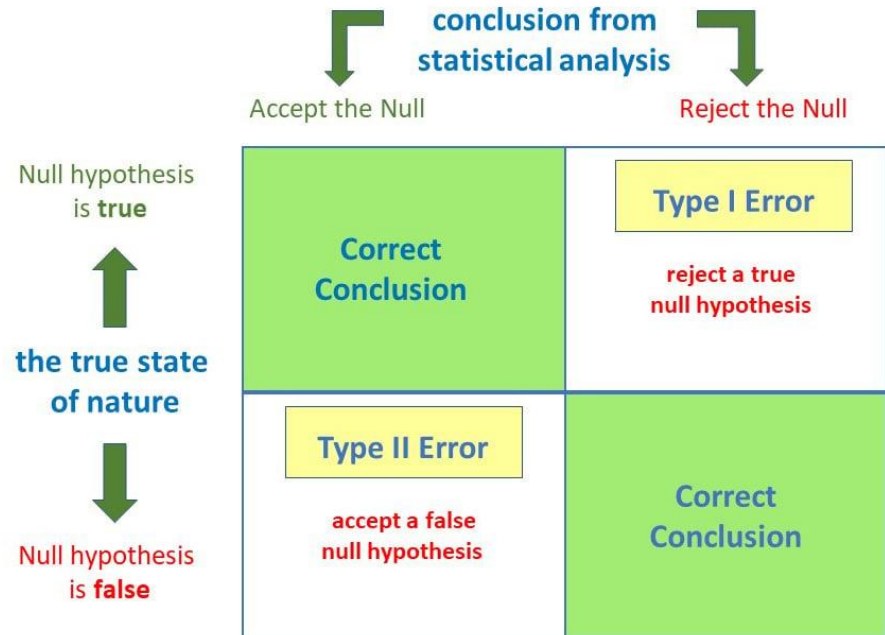
Step 4. Decide whether to accept or reject the null hypothesis.

Step 5. Draw a conclusion.

Type I and Type II error

- If H_0 is true, the probability of mistakenly reject H_0 is thus α .
(called **Type I Error**)

- If H_0 is not true, but we do not reject it, we make **Type II Error**.
The probability depends on the true population distribution.



Types of tests on mean

The form of the test (and thus the test statistic) depends on:

- The number of samples: one or two (this week) or more (Week 7)
- Which population parameter is of interest (the mean or proportion)

This week, We will focus on testing the mean(s):

- **One sample t-test/Z-test**
- **Paired samples t-test/Z-test**
- **Two independent samples t-test/Z-test**

For proportions: (only for large sample)

- **One sample Z-test**
- **Two independent samples Z-test**

Difference of Z-test and t-test on mean

- The difference lies in how the standard error of the mean (SEM) is calculated.
- We standardize the difference between the estimate and the population parameter by dividing by SEM. This takes both the population variance and sample size into consideration.
- When sample size is **small** AND population **variance is unknown**.
 - $SEM = s/\sqrt{n}$
 - The null distribution is t-distribution (**t-test**)
- When sample size is **large** OR population **variance is known**.
 - $SEM = \sigma/\sqrt{n}$
 - The null distribution is Normal-distribution (**Z-test**)

Conclusions

Basic concepts of hypothesis testing:

- How to state a Null hypotheses and alternate hypotheses?
- Why is the test statistic needed?
- Two approaches: Critical value approach and p-value approach
- What are Type I and Type II errors?

Reading: Oakshott Ch9, Taylor Ch7

Next we will consider:

- **One sample t-test/Z-test**
- **Paired samples t-test/Z-test**

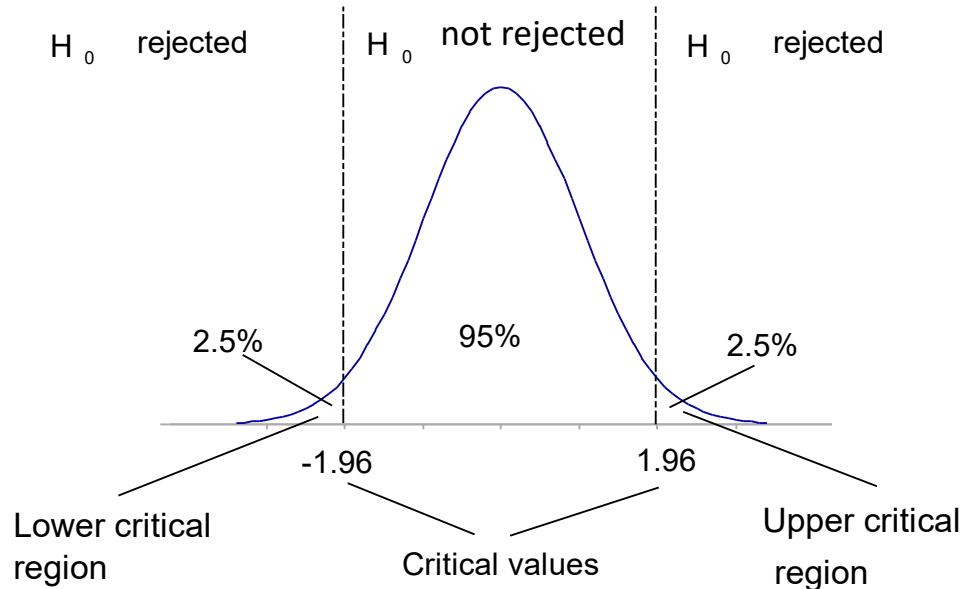
One sample Z-test for mean

- The test statistic is

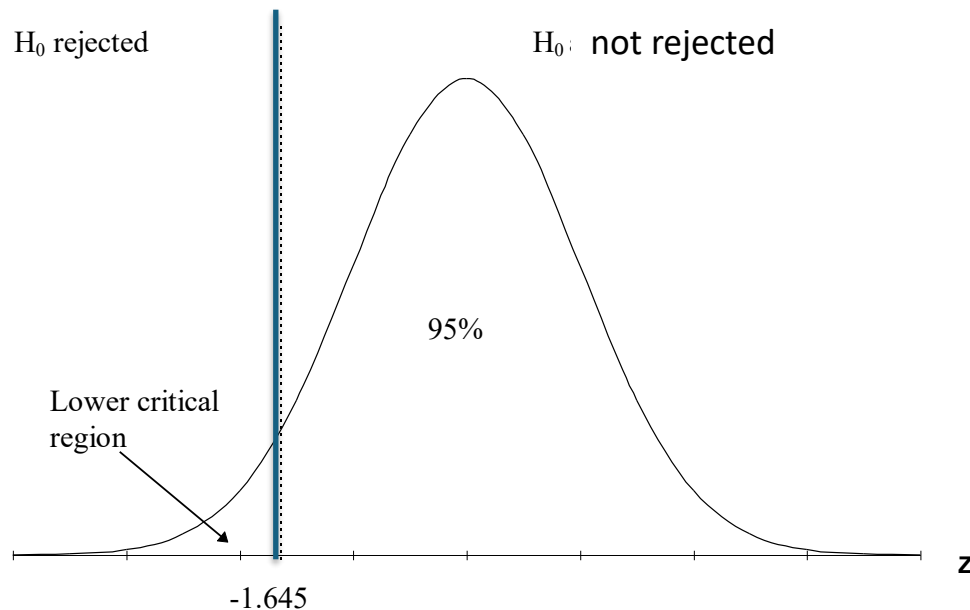
$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- The null distribution of Z is **N(0,1)**.
 - When the sample size is large we use the normal approximation (CLT)
 - The population distribution can be any distribution.
- The test can be one or two-sided.
 - Leading to different critical values/p-values.

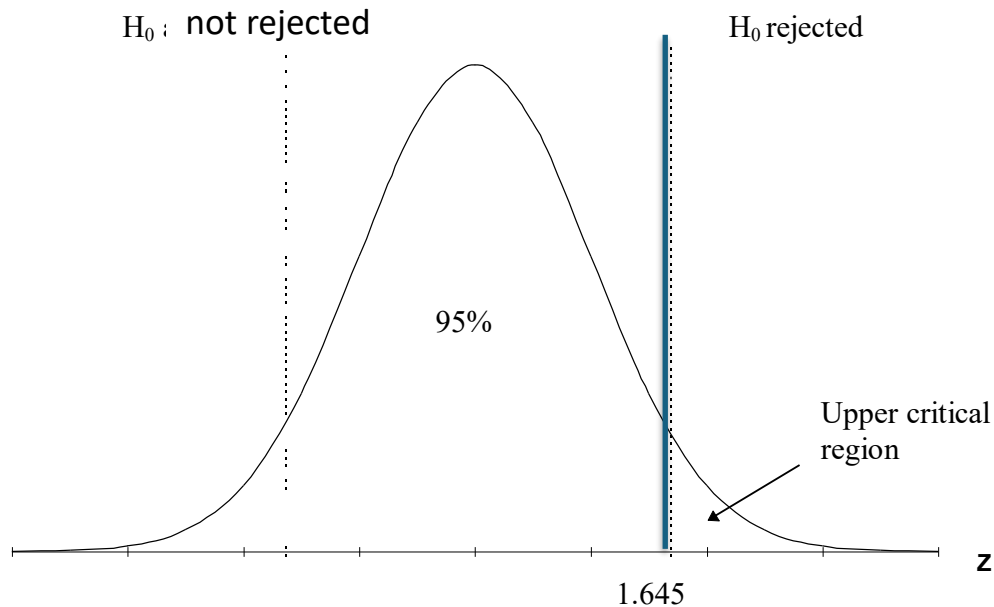
A two tailed test (not equal to) using a 5% significance level



One tailed test (less than) using a 5% significance level



One-tailed test (greater than) using a 5% significance level



Example: salt in pasty

- You care about the salt you eat. The manufacturer says each pasty contains, on average, 2.65g of salt. You test 10 and find that they contain an average of **2.8g**.
- Obviously, this is more than **2.65g**.
- Assuming $\sigma = 0.3g$, is it significantly more than 2.65g?



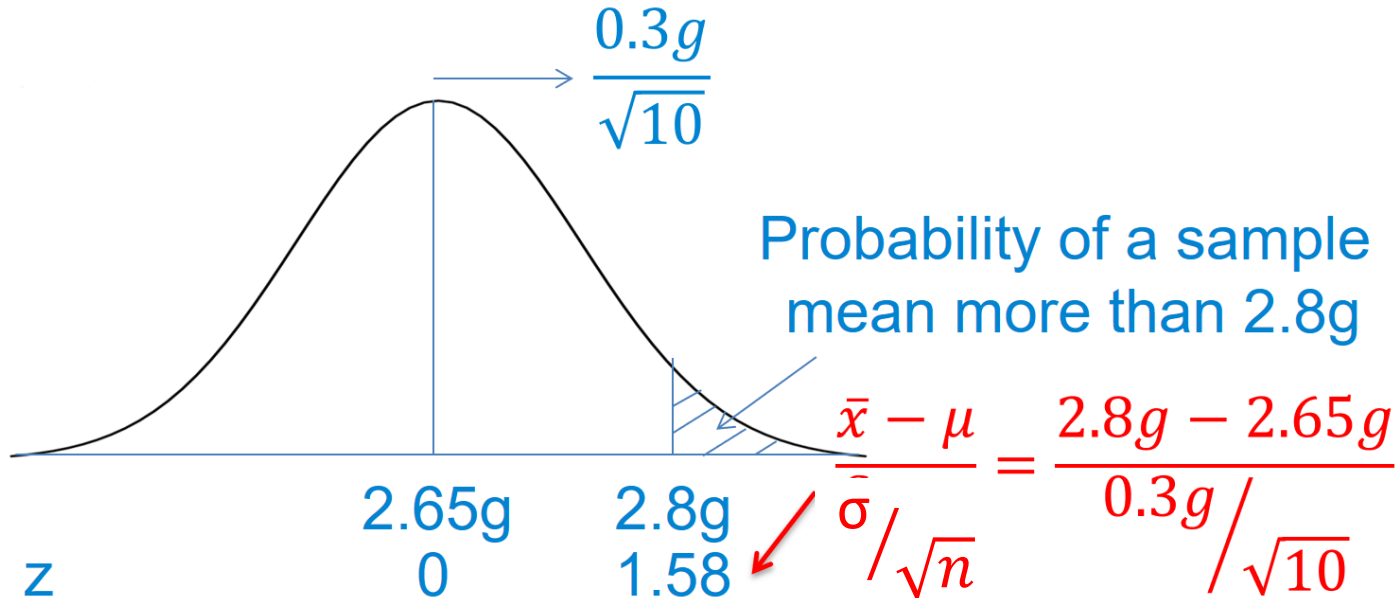
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We should use t-test if assuming $s=0.3g$.

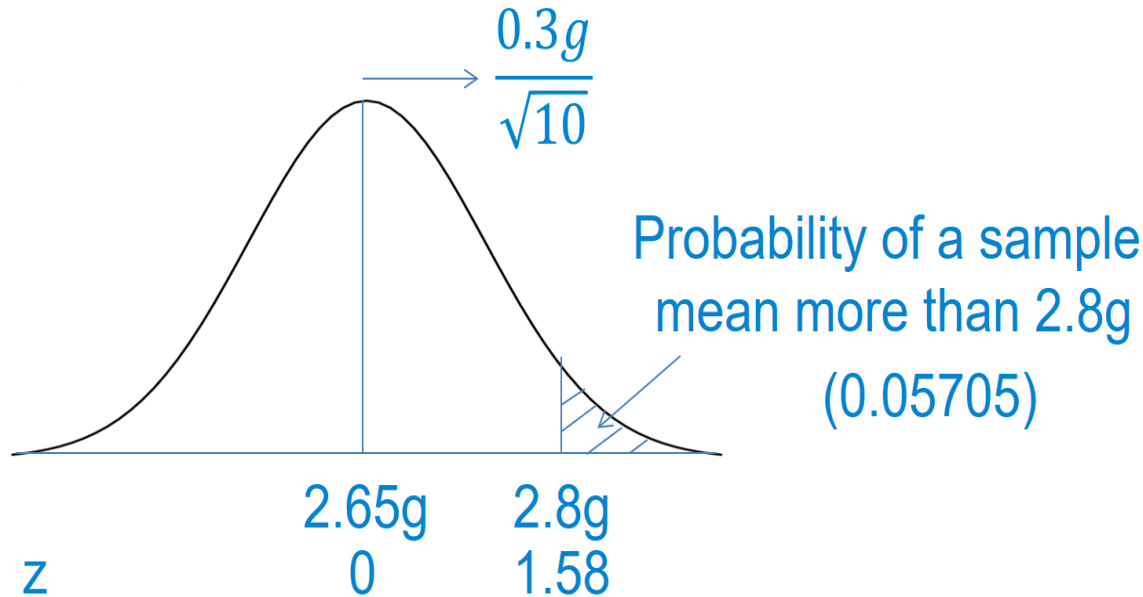
Using p-value (1)

There is now a z-row, with 0 for μ , and 1.58 under $\bar{x}$ (2.8g).

This is found by standardizing data, i.e. doing $(2.8g - 2.65g)$ over $(0.3g \text{ over } \sqrt{10})$.



Using p-value (2)



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.10	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.20	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.30	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.40	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.50	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.60	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.70	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.80	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.90	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.00	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.10	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.20	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.30	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.40	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.50	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408

Using p-value (3)

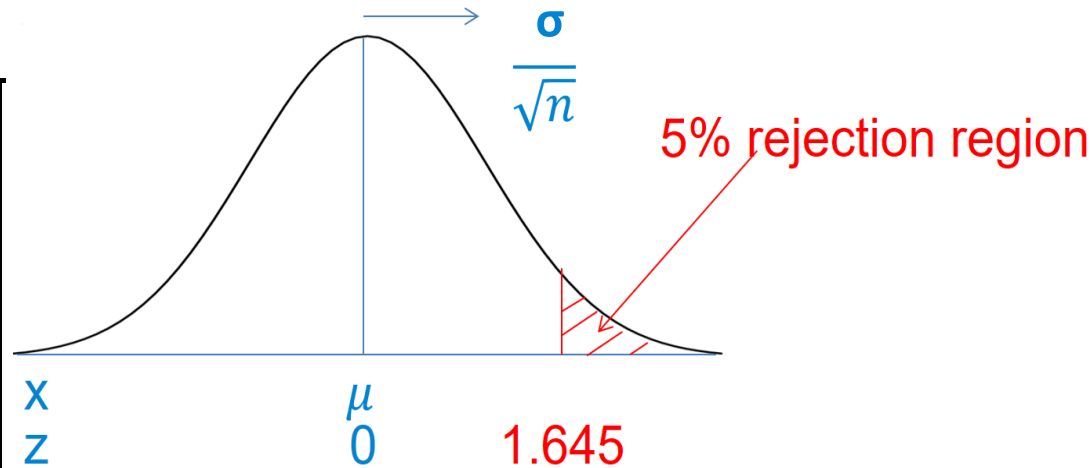
- Is a probability of 0.05705 low enough to legitimately conclude:
“Compared with 10%, that is unlikely. Assuming our sampling was representative, I do not believe their claim of 2.65g! The true result is clearly higher!”

- “Compared with 5%, well, the evidence is insufficient!”

Using Critical Values (1)

Find which value of z represents 5%, and compare our standardised $\bar{x}$ to it.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.10	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
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0.30	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.40	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.50	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.60	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.70	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.80	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.90	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.00	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.10	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.20	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.30	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.40	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.50	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.60	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449

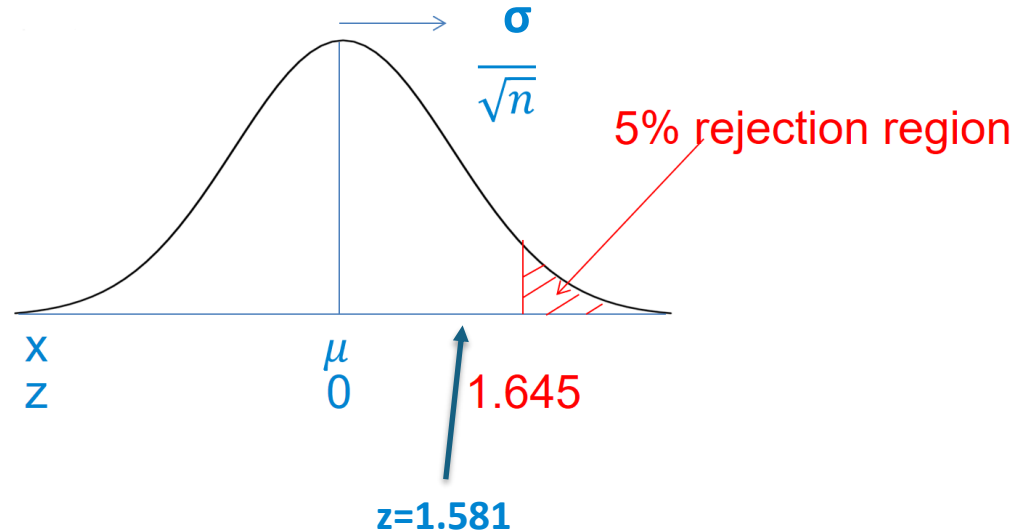


A normal distribution with mean μ (standardising to 0) and standard error s over root n . The area to the right of 1.645 is shaded and labelled the 5% rejection region.

Using Critical Values (2)

If $\bar{x}$ gives a $z > 1.645$, then the probability of an outcome bigger than $\bar{x}$ is **less** than 5%.

If $\bar{x}$ gives a $z < 1.645$, then the probability of an outcome bigger than $\bar{x}$ is **more** than 5%.



(One Tailed) Z-test

Null Hypothesis: $\mu = 2.65g$.

Alternate Hypothesis: $\mu > 2.65g$.

Significant level: 5%

Null distribution: $N(0,1)$

Critical Value: 1.645. (Upper bound)

Test Statistic:
$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{2.8g - 2.65g}{0.3g / \sqrt{10}} = 1.581.$$

As $z < 1.645$ (not in reject region), the null hypothesis is not rejected. We conclude that we **do not have sufficient evidence to reject** the original claim of 2.65g.

(Most presentations of the analysis will include the p-value, for us it was 0.05705.)

t-distribution

- In reality, we usually do not know σ , but we can calculate s .
- In such a case, we shouldn't use the normal distribution. (also $n=10$ is small)
- We should use a t-distribution instead. (Rather than a z-test, it would have been called a t-test.)
- With one-tail, 5% significance, null distribution is t distribution with **degree of freedom** = $n - 1 = 9$.
- We would have found a Critical Value of 1.833. (*The t-test is more conservative than the z-test because its distribution has heavier tails, requiring a larger test statistic to achieve the same level of statistical significance.*)
- Our Test Statistic (1.581) is still less than this.



PERCENTAGE POINTS OF THE T-DISTRIBUTION

One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073

(One Tailed) t-test

Null Hypothesis: $\mu = 2.65g$.

Alternate Hypothesis: $\mu > 2.65g$.

Significant level: 5%

Null distribution: $t(9)$

Critical Value: 1.883. (Upper bound)

Test Statistic:
$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{2.8g - 2.65g}{0.3g / \sqrt{10}} = 1.581.$$

As $t < 1.833$ (not in reject region), the null hypothesis is not rejected. We conclude that we **do not have sufficient evidence to reject** the original claim of 2.65g.

(Most presentations of the analysis will include the p-value, for us it was 0.07417.)

How to do two-sided test

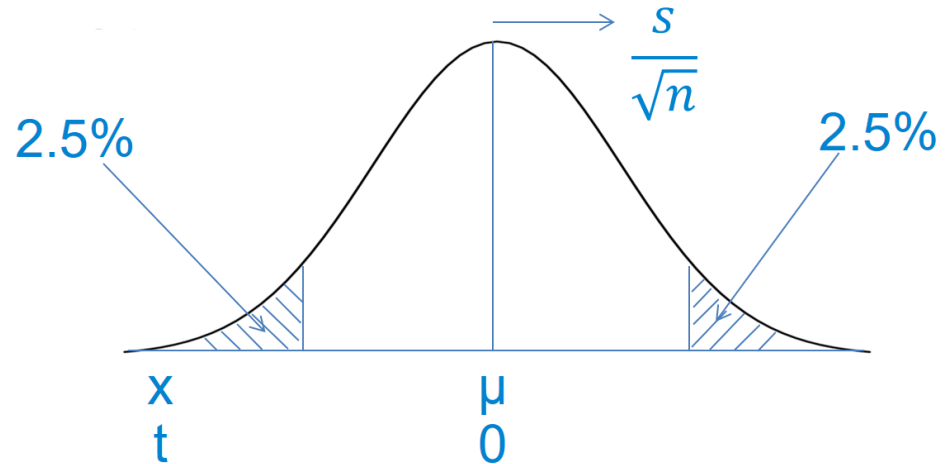
- We considered a sample moment being more than a hypothesized value.
- Looking at a sample moment being *less* than that works just the same (*shift the rejection region to the left, not the right*).

What if we want to consider if it “is it significantly different?”

- The test statistic is again calculated based on the parameter in the null hypothesis, thus is the same.
- That is a two-tail thing rather than a one-tail test (i.e. two CVs and **two rejection regions**, akin to a confidence interval, rather than a lower or upper bound.)

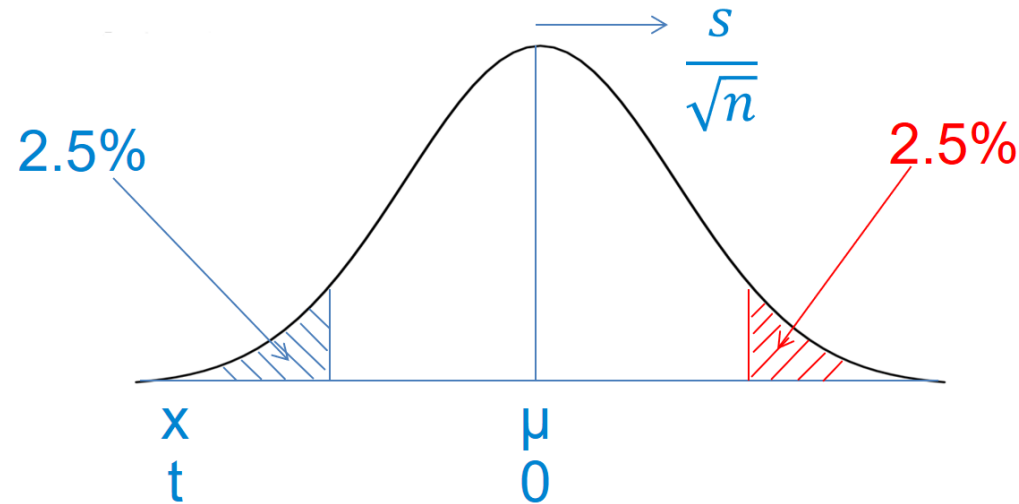
Two Tails – a diagram (1)

- We consider where our values would have to be for us to reject the null hypothesis.
- A diagram of a t-distribution (with mean and standard error generically labelled) with two regions marked, one at the end of each tail, each of size 2.5%.



Two Tails – a diagram (2)

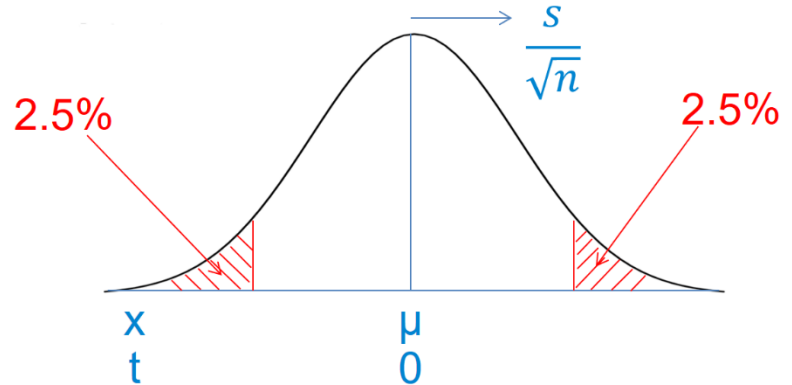
- Either a result too big or too small will allow us to reject the null.
- The higher region is highlighted in red to go with the phrase “too big”.



Two Tails – a diagram (3)

- We have two “rejection regions”.

Both regions are highlighted in red to go with the phrase “rejection regions”.



Which are two-tail situations?

- A. Is it what we expect, or not?
- B. Is it significantly greater than 2?
- C. Is it significantly different from 15?
- D. Is it significantly under 30?
- E. Are the two groups the same?

Confidence Intervals and Hypothesis Tests

Confidence Intervals use the sample to show where the parameter is likely to be.
(Sample centred)



Hypothesis Tests consider how far away the sample is from the parameter, then decides if that's acceptable.
(Theory centred)

Test on Proportion

We will only consider test on proportion with large sample. (CLT, Z-test)

Null Hypothesis: $H_0: \pi = \dots$

Alternate Hypothesis: $H_1: \pi \neq \dots$

Null distribution: $N(0,1)$.

Test Statistic:
$$Z = \frac{p - \pi}{\sqrt{\pi(1-\pi)/n}}$$

And then compare the Test Statistic (TS) and Critical Value (CV).

The variance is known

Test Statistic:

$$Z = \frac{p - \pi}{\sqrt{\pi(1 - \pi)/n}}$$

Note: not the same as in confidence intervals!

- We are assuming we know π , so we can make use of it.
- The variance of the Bernoulli distribution is $\pi(1 - \pi)$.

Example: test on proportion

Is a result of 59% out of $n=95$ significantly different from 50%?

Null Hypothesis: $H_0: \pi = 0.5$

Alternate Hypothesis: $H_1: \pi \neq 0.5$

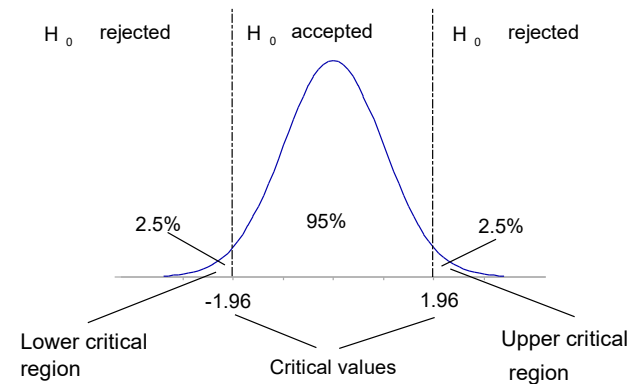
Null distribution: $N(0,1)$

Significant level: 5%, two-sided

Critical Value: -1.960 and 1.960

Test Statistic:

$$Z = \frac{p - \pi}{\sqrt{\pi(1-\pi)/n}} = \frac{\frac{59}{95} - 0.5}{\sqrt{0.5(1-0.5)/95}} = 2.360$$



TS > upper CV, so we conclude that 59% of 95 is significantly different from 50%.

Conclusions

- We understand the ideas behind hypothesis testing
- We know how to perform tests of hypothesis on the mean of a population
- We know how to perform tests of hypothesis on a percentage

Next we will consider tests for two populations:

- Paired samples t-test/Z-test
- Two independent samples t-test/Z-test

Exercise 1

A food manufacturer processes and cans baked beans. The net weight of a standard can of beans is supposed to be 420g but a random sample of 50 cans gave an average weight of 415g with a standard deviation of 30g. Is there any evidence that the true mean weight is not 420g? (Use a 5% significance level)

Solution

$$H_0 : \mu = 420\text{g} \quad H_1 : \mu \neq 420\text{g}$$

$$SEM = \frac{30}{\sqrt{50}}$$
$$= 4.243$$

$$Z = \frac{\bar{x} - \mu}{SEM}$$

$$Z = \frac{415 - 420}{4.243}$$
$$= -1.178$$

Conclusion of the test result

As the test statistic is between -1.96 and 1.96 we cannot reject H_0 . That is, there is no evidence at the 5% level of significance that the mean is not 420g

Exercise 2

A travel agent believes that half its clients will take the holiday insurance package that it has negotiated with an insurance company. In a random sample of 200 clients, 120 bought the package. Is there any evidence that the true percentage take-up is significantly different to 50%? (Use a 5% significance level)

Solution

This is a test of percentages

$$H_0 : \pi = 50\% \quad H_1 : \pi \neq 50\%$$

$$\begin{aligned}\text{STEP} &= \sqrt{\frac{\pi(100-\pi)}{n}} \\ &= \sqrt{\frac{50 \times 50}{200}} \\ &= 3.536\end{aligned}$$

$$\begin{aligned}P &= \frac{120}{200} \times 100 \\ &= 60\%\end{aligned}$$

$$\begin{aligned}Z &= \frac{60 - 50}{3.536} \\ &= 2.828\end{aligned}$$

Conclusion of the test result

As the test statistic is outside the region -1.96 to 1.96 we reject the null hypothesis and conclude that the true percentage take up is different to 50% at the 5% significance level. In fact, it appears that the take up is greater than 50%.

Exercise 3

A trade union is considering strike action and intends to ballot its large membership on the issue. In order to gauge the likely result of the ballot, a survey was conducted among a random sample of members. Of the 60 people surveyed, 36 were in favour of a strike. Would the ballot give the required (simple) majority for a strike at a 5% significance level

Solution

$$H_0: \pi = 50\%$$

$$H_1: \pi > 50\%$$

$$\begin{aligned}\text{STEP} &= \sqrt{\frac{50 \times 50}{60}} \\ &= 6.455\end{aligned}$$

$$\begin{aligned}P &= \frac{36}{60} \times 100 \\ &= 60\%\end{aligned}$$

$$\begin{aligned}Z &= \frac{60 - 50}{6.455} \\ &= 1.549\end{aligned}$$

Conclusion of the test result

The critical value at a 5% significance level is 1.645 and as the test statistic is less than this, we cannot reject H_0 . We conclude that there is no evidence that a majority would be in favour of a strike.

Some useful slides for practicals:

Computing and SPSS

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Height (m)	159	1.7061	.12689	.01006

One-Sample Test

Test Value = 1.68

	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Height (m)	2.597	158	.010	.02613	.0063	.0460

- An output from SPSS. There are two tables: One-Sample Statistics, covering the number of datapoints (159), mean (1.7061), standard deviation (0.12689) and STEM (0.01006), and the second, called One Sample Test. This contains: Test Value = 1.68, t (2.597), df (158), Sig (2-tailed) (0.010), Mean difference (0.02613), and 95% confidence interval of the difference, (0.0063 to 0.0460).

Computing and SPSS

One-Sample Statistics

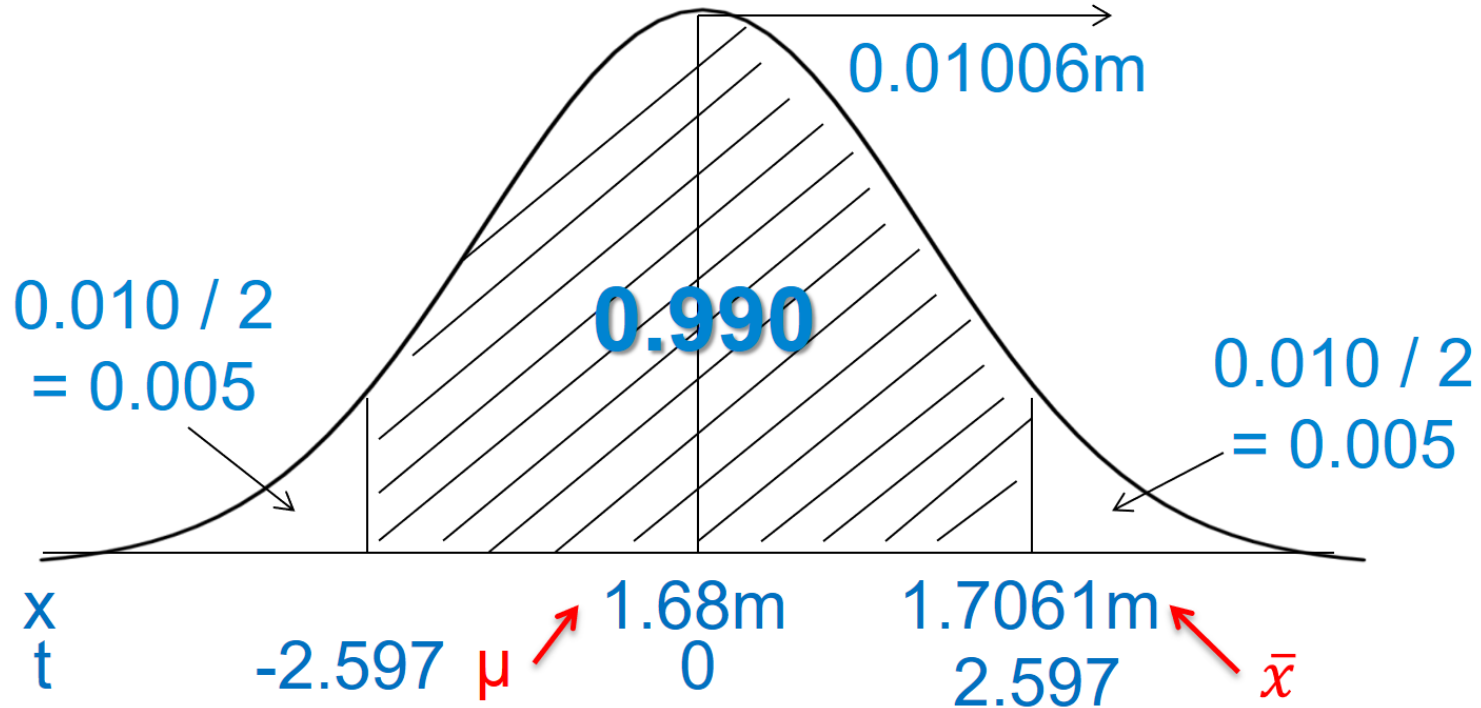
	n	$\bar{x}$	s	$\frac{s}{\sqrt{n}}$
	N	Mean	Std. Deviation	Std. Error Mean
Height (m)	159	1.7061	.12689	.01006

One-Sample Test

	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Height (m)	2.597	158	.010	.02613	.0063	.0460

$\frac{\bar{x} - \mu}{s/\sqrt{n}}$
 $n-1$
p-value
 $\bar{x} - \mu$
 $(\bar{x} - \mu) \pm CC \frac{s}{\sqrt{n}}$

More in-depth diagram



Basic Hypothesis Testing

Week 5 (Part 3)

From the last lecture: Confidence Intervals and Hypothesis Tests

Confidence Intervals use the sample to show where the parameter is likely to be.
(Sample centred)



Hypothesis Tests consider how far away the sample is from the parameter, then decides if that's acceptable.
(Theory centred)

From the last lecture: Confidence Intervals and Hypothesis Tests

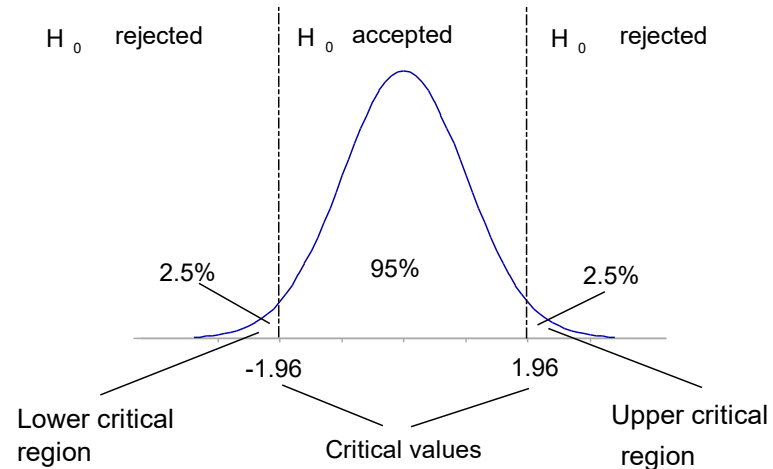
Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally, 5%), null distribution, and determine the critical values

Step 3. Calculate the test statistic

Step 4. Decide whether to accept or reject the null hypothesis.

Step 5. Draw a conclusion.



Plan of the session

Hypothesis test involving two means

- Paired samples t-test/Z-test
- Two independent samples Z-test
- Two independent samples t-test

Difference between two continuous variables – Paired data (1)

- In our sample data, we have both people's heights, AND their guesses for what the average height is.
- We might wonder whether people's guess of average height is biased or not.

Height	Guess
1.56	1.70
1.78	1.72
1.54	1.68
1.70	1.70
1.70	1.71
1.83	1.73
1.73	1.73
...	...

Difference between two continuous variables – Paired data (2)

- Take the differences between each persons guess and their actual height.
- And then compare this difference to a hypothesized value of 0.
- Then one can implement a single sample test for the mean of the differences.

Height	Guess	Difference
1.56	1.70	0.14
1.78	1.72	-0.06
1.54	1.68	0.14
1.70	1.70	0.00
1.70	1.71	0.01
1.83	1.73	-0.10
1.73	1.73	0.00
...	...	...

Paired samples

- The null hypothesis is that **the *difference of the population means is zero*** and the alternative hypothesis can be either one or two-tailed.
- The test statistic for this test is:

$$t = \frac{\bar{x}_d - 0}{s_d / \sqrt{n}}$$

- The null distribution is $t(n-1)$ if the sample size is small, and is $N(0,1)$ if the sample size is large. (The same as one sample test)
- Use critical value approach or p-value approach to make the decision.

Example of SPSS output

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Height (m)	1.7061	159	.12689	.01006
	Class Mean Height Guess (m)	1.7207	159	.08219	.00652

Paired Samples Test

		Paired Differences							
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	Sig. (2-tailed)
Pair 1	Height (m) - Class Mean Height Guess (m)	-.01460	.15326	.01215	-.03861	.00940	-1.201	158	.231

Example of SPSS output

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Height (m)	1.7061	159	.12689	.01006
	Class Mean Height Guess (m)	1.7207	159	.08219	.00652

Paired Samples Test

		Paired Differences			95% Confidence Interval of the Difference			
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df
Pair 1	Height (m) - Class Mean Height Guess (m)	-.01460	.15326	.01215	-.03861	.00940	-1.201	158

$$\frac{\bar{x}_d - 0}{s_d / \sqrt{n}}$$

$$\bar{x}_d$$

$$s_d$$

$$s_d / \sqrt{n}$$

$$s_d / \sqrt{n}$$

$$n - 1$$

Independent Samples

- Paired data is when you have two comparable datapoints per person (their height and their guess of their height, for example).
- Consider a case of two **independent** groups of people, with one measurement each.
- You may not have the direct association between datapoints, and so can't calculate the difference.

What would this look like?

For example, in our sample dataset, we could be interested in whether **Gender** (a dichotomous variable) has an association with **Height** (a continuous variable).

We don't have a link between people in two gender groups to just take differences.

Height	Gender
1.56	F
1.78	M
1.54	F
1.70	M
1.70	F
1.83	M
1.73	F
...	...

Independent samples Z-test (1)

- If we know the population standard deviations (rare) or have big enough samples (more common), we can use the following test statistic:

$$TS = \frac{\overline{x_1} - \overline{x_2}}{STED}$$

$$\text{where } STED = \sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}$$

CV from $N(0,1)$ table.

(Note $STED^2 = STEM_1^2 + STEM_2^2$.)
(If we don't know σ we just use s .)

Independent samples Z-test (2)



A Researcher is investigating whether two credit cards' customers have different spending patterns.

Do the consumers have significantly **different mean** spending habits?

The **149** Amber customers sampled had a mean spend of **£55** with a standard deviation of **£4**. The **135** Burlow customers sampled had a mean spend of **£48** with a standard deviation of **£5**.

Independent samples Z-test (3)

- A Researcher is investigating whether two credit cards' customers have different spending patterns.
- Do the consumers have significantly **different mean** spending habits?
- The **149** Amber customers sampled had a mean spend of **£55** with a standard deviation of **£4**. The **135** Burlow customers sampled had a mean spend of **£48** with a standard deviation of **£5**.

$$TS = \frac{\bar{x}_1 - \bar{x}_2}{STED}$$

$$\text{where } STED = \sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}$$

CV from N(0,1) table.

(Note $STED^2 = STEM_1^2 + STEM_2^2$.)
(If we don't know σ we just use s .)

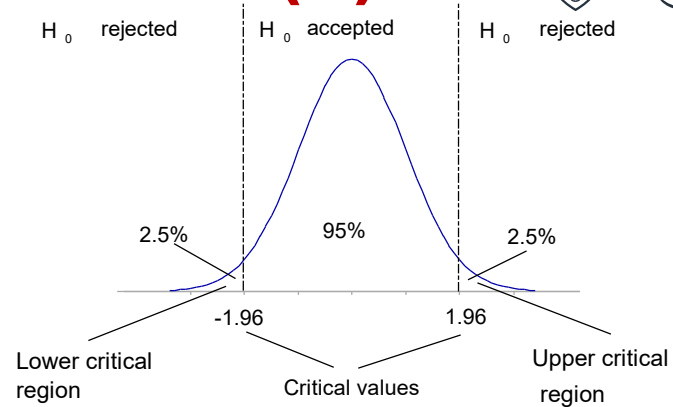
Independent samples Z-test (4)

Null Hypothesis: $H_0: \mu_1 = \mu_2$

Alternate Hypothesis: $H_1: \mu_1 \neq \mu_2$

$$STED = \sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)} = \sqrt{\left(\frac{£4^2}{149} + \frac{£5^2}{135}\right)} = \sqrt{(£0.33^2 + £0.43^2)} = £0.54$$

$$TS = \frac{\bar{x}_1 - \bar{x}_2}{STED} = \frac{£55 - £48}{£0.54} = 12.942$$



Critical Value: 2 tails (different), 5% (not mentioned), so CV = -1.960 and 1.960.

TS > upper CV, so yes, Amber and Burlow customers spend significantly different amounts, on average.

Independent samples t-test (1)

If we don't know σ_1 and σ_2 **and** don't have big enough samples... it's a little more convoluted. ***We need to assume the two samples come from populations with the same standard deviation.***

We estimate this “true” standard deviation by *pooling* the sample standard deviations:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

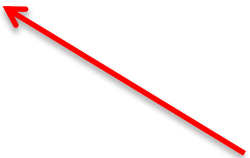
- This is just a weighted average of each sample's standard deviation.
- The **degrees of freedom** is $n_1 + n_2 - 2$.

Independent samples t-test (2)

- Everything else stays the same:

$$STED = \sqrt{\left(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}\right)} = s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$TS = \frac{\bar{x}_1 - \bar{x}_2}{STED}$$



(This presentation makes it easier on the calculator!)

The Critical Value is from t-distribution with **df** = $n_1 + n_2 - 2$.

Example: Two conveyor belts (1)



Two conveyor belts of cans are meant to have the same weights.
Is there a significant difference between the cans' weights?

Small samples are taken:

- 16 cans are taken (sampled) from one belt (with mean 200g and sd 12.3g)
- But only 6 cans are sampled from the other (with mean 205g and sd 10g).

Example: Two conveyor belts (2)



A rule of thumb is if you have to look at the infinity row, you have large samples. If not, they are small.

Here, it's row $16+6-2=20$, small samples. That means we have to do more work.

What are the hypotheses?

Null Hypothesis: No difference. That is, $H_0: \mu_1 = \mu_2$.

Alternate Hypothesis: Some difference. $H_1: \mu_1 \neq \mu_2$.

Example: Two conveyor belts (3)



Pooled standard deviation:

$$\begin{aligned} s_p &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \\ &= \sqrt{\frac{(16 - 1)12.3g^2 + (6 - 1)10g^2}{16 + 6 - 2}} = 11.767g \end{aligned}$$

(Note how this is between the 12.3g and 10g, but closer to 12.3g, which is from the bigger sample.)

Example: Two conveyor belts (4)



And the standard error:

$$STED = s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} = 11.767g \sqrt{\frac{1}{16} + \frac{1}{6}} = 5.633g$$

And the Test Statistic:

$$TS = \frac{\bar{x}_1 - \bar{x}_2}{STED} = \frac{200g - 205g}{5.633g} = -0.888$$

Example: Two conveyor belts (5)



The **Critical Value**:

$$df = n_1 + n_2 - 2 = 16 + 6 - 2 = 20,$$

2 tails, 5% significance, gives CV=-2.086 and 2.086.

TS (0.888) is between -2.086 and 2.086, so there is **no** significant difference, and the weights of cans from different belts are **not** noticeably different.

(this is a dichotomous associated with a scalar variable example.)

Independent small samples – SPSS

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Height (m)	Male	64	1.8056	.09565	.01196
	Female	95	1.6391	.09826	.01008

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
Height (m)	Equal variances assumed	.016	.899	10.591	157	.000	.16651	.01572	.13546	.19757
	Equal variances not assumed			10.647	137.755	.000	.16651	.01564	.13559	.19744

Independent small samples – SPSS

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Height (m)	Male	64	1.8056	.09565	.01196
	Female	95	1.6391	.09826	.01008

Then one of these. Top one if the variances **aren't** significantly different, bottom one otherwise.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means					
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference
									Lower Upper
Height (m)	Equal variances assumed	.016	.899	10.591	157	.000	.16651	.01572	.13546 .19757
	Equal variances not assumed			10.647	137.755	.000	.16651	.01564	.13559 .19744

Here first. Are the variances significantly different?

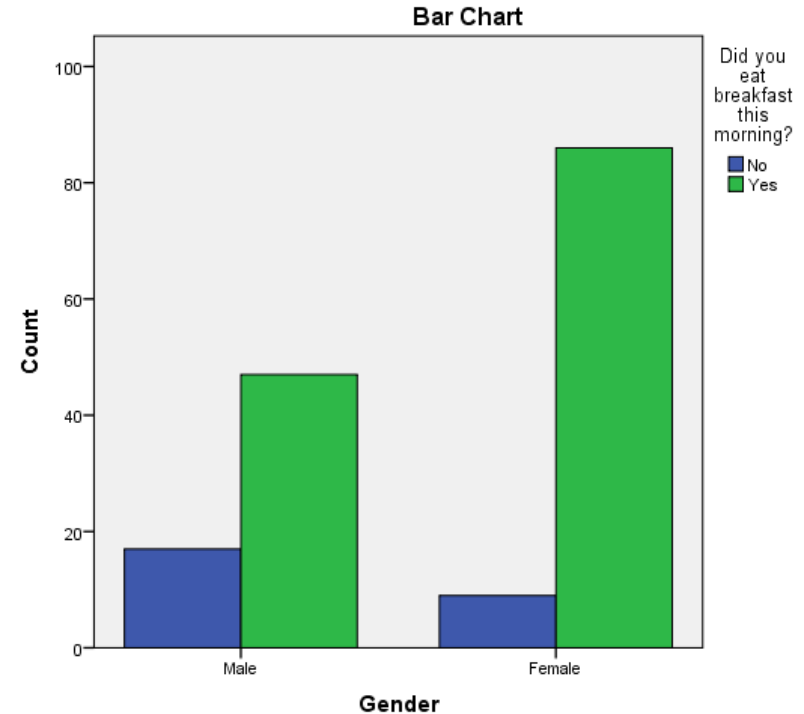


Lastly... dichotomous vs dichotomous

Gender * Did you eat breakfast this morning?
Crosstabulation

Count

		Did you eat breakfast this morning?		
		No	Yes	Total
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159



Two independent samples Z-test on proportions

Start with a similar null hypothesis; which of the two groups you're in does NOT affect the percentage / proportion.

Null Hypothesis: No difference. That is, $H_0: \pi_1 = \pi_2$.

Alternate Hypothesis: Some difference. $H_1: \pi_1 \neq \pi_2$.

By assuming $\pi_1 = \pi_2$, the two measures of spread are the same as well.
So that implies pooling again.

We will see an example in the next week.

Conclusions

We considered Hypothesis Testing. We learned about:

- Null hypotheses
- Alternate hypotheses
- Critical values
- P-values
- Test statistics

For means:

One sample t-test/Z-test

Two paired samples t-test/Z-test

Two independent samples t-test/Z-test

For proportions: (only for large sample)

One sample Z-test

Two independent samples Z-test

Next week: The Chi-square Test